WEATHER IMAGE CLASSIFICATION

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DATA SCIENCE

INTRODUCTION

- Weather classification using image data
- Based on Deep Learning (CNN)
- Classifies sunny, cloudy, rainy images
- Learns visual weather patterns
- Useful for real-world applications
- Demonstrates AI in weather analysis

OBJECTIVES

- To classify weather images using CNN
- To identify weather types (sunny, cloudy, rainy)
- To automate weather recognition from images
- To evaluate classification accuracy
- To apply deep learning in weather analysis

KEY FEATURES

- CNN-based weather image classification
- Automatic feature extraction from images
- Multi-class weather prediction
- Image preprocessing and training pipeline
- Model accuracy evaluation
- Real-world application in weather analysis

DATASET DESCRIPTION

- Weather image dataset containing different weather conditions
- Classes include sunny, cloudy, rainy, etc.
- Images collected from open-source sources
- Labeled by weather category
- Used for training and testing the CNN model

SAMPLE TRAINING IMAGE

CONCLUSION

- CNN model successfully classifies weather images
- Achieves good accuracy in prediction
- Automatically learns weather features
- Useful for real-world weather recognition
- Demonstrates effectiveness of deep learning

THANKYOU…